

Martin Fournier

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Research interests : *Intracluster medium, Computational astronomy
Astrophysical fluid dynamics*

EDUCATION

PhD Student in Astrophysics

FROM JUL. 2023

University of Hamburg, with Prof. Marcus Brüggen and Dr. Philipp Grete

HAMBURG, GERMANY

Master Degree in Astronomy & Astrophysics

SEP. 2022 – JUL. 2023

Observatoire de Paris & Université Paris-Cité

PARIS, FRANCE

- Master Thesis : “Cosmic rays propagation in the Milky-Way”, ENS Lyon
- Advisors : Dr. Jérémy Fensch and Dr. Benoit Commerçon

Master Degree in Physics and Engineering

SEP. 2019 – JUN. 2022

Phelma – Grenoble INP – Grenoble Alpes University

GRENOBLE, FRANCE

- Master Internship : “Properties of Magellanic Clouds Analogs in TNG50”, Max-Planck-Institut für Astronomie
- Advisor : Dr. Annalisa Pillepich

PUBLICATIONS

First author publications:

- M. Fournier et al., XMAGNET – Stir before serving: a Lagrangian perspective on mixing-driven condensation in the intracluster medium, 2026, *Submitted to A&A*
- M. Fournier et al., XMAGNET: Velocity structure functions of active galactic nucleus-driven turbulence in the multiphase intracluster medium, 2025, *Published, A&A* (cited: 8)
- M. Fournier et al., The properties of magnetised cold filaments in a cool-core galaxy cluster, 2024, *Published, A&A* (cited: 22)
- M. Fournier et al., Past activity of Sgr A* is unlikely to affect the local cosmic-ray spectrum, 2023, *Published, A&A*

Co-author publications:

- O. Piana, ..., M. Fournier et al., BlackHoleWeather – Spin-coupled chaotic cold accretion across the meso scale: Morphology and thermodynamics, 2026, *Submitted to A&A*
- V. Cammelli, ..., M. Fournier et al., BlackHoleWeather – Jet-regulated chaotic cold accretion across the meso scale: Morphology and thermodynamics, 2026, *Submitted to A&A*
- F. Barbani, ..., M. Fournier et al., BlackHoleWeather – Chaotic cold accretion across the meso-scale: variability and kinematics, 2026, *Submitted to A&A*

- F. Barbani, ..., **M. Fournier** et al., BlackHoleWeather – Chaotic cold accretion across the meso-scale: morphology and thermodynamics, 2026, *Accepted, A&A*
- D. Prasad, ..., **M. Fournier** et al., XMAGNET : Kinetic, Thermal and Magnetic AGN Feedback in Massive Galaxies at Halo Masses $\sim 10^{13.5} M_{\odot}$, 2025, *Published, MNRAS*
- P. Grete, ..., **M. Fournier** et al., The XMAGNET Exascale MHD simulations of SMBH feedback in galaxy groups and clusters: Overview and preliminary cluster results, 2025, *Published, ApJ* (cited: 8)
- A. Pillepich, ..., **M. Fournier** et al., Milky Way and Andromeda analogs from the TNG50 simulation, *Published, MNRAS* (cited: 111)

COMPUTING PROJECTS

- **Co-PI** of “*Magnetohydrodynamical simulations of jet-cloud interactions in the intracluster medium*”, **2 Mcore-h on JUWELS GPU Booster** (Jülich Supercomputing Centre), **2024-2026**
3 Mcore-h on JUPITER GPU Booster (Jülich Supercomputing Centre), **2026-2027**
- **Project contributor** for “*Feedback and energetics from magnetized AGN jets in galaxy groups and clusters*”, **80 Mcore-h on Frontier** (Oak Ridge Leadership Computing Facility), **2023-2026**

TALKS

Invited talks

- **BlackHoleWeather: Toward a Unified Theory of Feeding and Feedback**, *Sexten*, 2025

Contributed talks

- **8th ICM Theory and Computation Workshop***, *Brno Observatory and Planetarium*, 2026
- **Modelling of Multiphase Astrophysical Media**, *Schloss Ringberg*, 2025
- **MIST2025 – Cosmic turbulence and Magnetic fields**, *Cargèse*, 2025
- **Journées de la Société Française d’Astronomie & d’Astrophysique (SF2A)**, *Toulouse*, 2025
- **Galaxy Clusters & Radio Relics II**, *Cambridge, MA*, 2024
- **Journées de la Société Française d’Astronomie & d’Astrophysique (SF2A)**, *Marseille*, 2024

*Poster talk

Colloquia and Seminars talks

Harvard–Smithsonian CfA, 2024 & 2025; Observatoire de Paris, 2025, 2026; Max–Planck–Institut für Meteorologie, 2025; Princeton University, 2025; Newcastle University, 2024; Institut für Theoretische Astrophysik, 2024; Max–Planck–Institut für Astronomie, 2024; École Normale Supérieure de Lyon, 2024

TEACHING & PROFESSIONAL SERVICE

- **Teaching Assistant** for “*Active Galactic Nuclei and Supermassive Black Holes*”, Hamburg Universität, 12 hours
- **Lecturer** for “*Introductory Python for Physicists*”, Hamburg Universität, 23 hours
- **Referee** for *Astronomy & Astrophysics* (A&A)

OUTREACH

- **Main organizer** of a sold-out Science Slam show at the Hamburg Planetarium with over 250 attendees.
- **Speaker** in various public outreach events, including popular science talks and Science Slams.
- **Production of videos** based on the IllustrisTNG simulations, featured on ARTE and *When Fish Began to Crawl* (2023), a film on climate change by BAFTA-winning director Morgan McKinnon, premiered at the Edinburgh Science Festival.

TECHNICAL SKILLS

Languages : Fortran, Python, C++, L^AT_EX
Frameworks & Codes : KOKKOS, AREPO, RAMSES, ATHENAPK